

Problem 3.1

Solution:

(i) Subproblems:

Define $C(u, i)$ as the maximum number of leaves that can be selected in the subtree rooted at u , such that selected leaves are at least distance k apart, and every selected leaf is at least distance i from u . Want $C(\text{root}, 0)$.

(ii) Recursive formula:

Let L and R be the left and right children of u . For leaf nodes (base case):

$$C(u, i) = \begin{cases} 1, & \text{if } i = 0 \\ 0, & \text{if } i > 0 \end{cases}$$

For internal nodes:

$$C(u, i) = \max_{\substack{0 \leq l, r \leq k+1-i \\ 2i+l+r \geq k}} \{C(L, i-1+l) + C(R, i-1+r)\}$$

Since $C(u, i + \Delta) \leq C(u, i)$ for $\Delta > 0$, $C(u, i)$ only takes $O(k)$ time to compute.

(iii) Evaluation order:

Use a bottom-up approach via postorder traversal. Compute C values for child nodes before their parent.

(iv) Time and space complexity:

Each node has $O(k)$ states, and each state is computed in $O(k)$ time.

Total time complexity: $O(nk^2)$

Total space complexity: $O(nk)$